

Name: _____ Date: _____

Period: _____

Primary Objective

By the end of this lesson, I will be able to...

1. Identify an appropriate function type for a data set or context by analyzing _____ of change. (LO 1.13.A)
2. Describe the _____ and _____ restrictions embedded in a function model. (LO 1.13.B)

Vocabulary & Key Concepts

first differences:

second differences:

domain restriction:

piecewise-defined function:

model assumption:

Hook

“A scientist hands you a table of mystery data. How do you decide which function type to use?”

Without graphing, you can inspect the _____ to detect the function’s degree.

Rate-of-Change Clues: The Difference-Table Method

Setup: Inputs must be _____-spaced. Compute differences of the output column.

x	y	Δy (1st diff.)	$\Delta^2 y$ (2nd diff.)
0	1	—	—
1	4	_____	—
2	11	_____	_____
3	22	_____	_____
4	37	_____	_____

Decision tree:

- 1st differences **constant** $\Rightarrow$ model is _____ (degree _____)
- 2nd differences **constant** $\Rightarrow$ model is _____ (degree _____)
- 3rd differences **constant** $\Rightarrow$ model is _____ (degree _____)
- n th differences constant $\Rightarrow$ model is degree-_____ polynomial

Key idea: Constant *first* differences mean a _____ rate of change. Constant *second* differences mean the rate of change is itself _____.

Shape and Context Clues

When you have a description but no table, use context:

Context clue	Best model type
Roughly constant rate of change	_____
Area or 2-D measurement	_____
Volume or 3-D measurement	_____
Multiple local maxima/minima	_____
Behavior changes at a threshold	_____

Special rule: A degree- n polynomial can model a data set with exactly _____ points (with distinct inputs).

Piecewise Models

A piecewise-defined function uses _____ formulas over _____ intervals.

Example scenario: A delivery driver travels at a constant speed for the first 20 minutes, stops for 5 minutes, then accelerates home. This situation requires _____ pieces because the _____ changes over different intervals.

To describe a piecewise model, state:

1. The _____ for each piece.
2. The type of function for each piece (e.g., _____, _____).
3. The _____ conditions (where each piece applies).

Stating Assumptions and Restrictions

Every model embeds assumptions. Fill in the sentence frame:

“This model assumes that _____ remains _____ over
_____. The domain is restricted to _____ because
_____.”

Example: A quadratic models the area A of a rectangle with a fixed perimeter of 120 ft, where ℓ is one side length.

Assumption: _____

Domain restriction: _____

Range restriction: _____